Progress Report

Multi-Agent Reinforcement Learning for Adaptive Traffic Signal Optimization in Smart Cities

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1. Overview

This report summarizes the current progress of the research project focused on developing a **Multi-Agent Reinforcement Learning (MARL)** framework for adaptive traffic signal optimization.

The project is currently in the **foundational and system-building phase**, with emphasis on establishing a reliable simulation environment and baseline for future experimentation.

2. Work Completed

2.1 Simulation Environment

A custom **grid-based traffic simulation environment** has been successfully developed.

Key Features:

- Configurable intersection grid (initially 4×4)
- Multi-directional traffic flow (N, S, E, W)
- Discrete-time simulation model
- Vehicle generation with stochastic patterns
- Queue formation and movement logic

This environment serves as the **core experimental platform** for all subsequent research activities.

2.2 Baseline Traffic System

A **fixed-time traffic signal control system** has been implemented.

Characteristics:

- Predefined signal phases
- Fixed duration cycles
- Deterministic behavior

Purpose:

- Establish a benchmark for comparison
- Validate simulation correctness

- Provide reference metrics

2.3 Metrics & Evaluation Framework

An initial evaluation framework has been developed to measure system performance.

Metrics Implemented:

- Average waiting time
- Queue length
- Throughput
- Congestion indicators

Additional Features:

- Structured logging of simulation data
- Support for repeated experimental runs

2.4 Visualization Dashboard

A basic visualization layer has been implemented to monitor the system.

Capabilities:

- Real-time traffic simulation view
- Signal state visualization
- Preliminary metric display

This enables **intuitive observation of traffic behavior** and system performance.

3. Current Work (In Progress)

3.1 MARL System Design

The project is currently transitioning into the **learning-based control phase**.

Ongoing Activities:

- Defining agent architecture per intersection

- Designing state, action, and reward representations
- Selecting suitable reinforcement learning algorithms
- Structuring training workflow

3.2 Initial Integration Planning

Work is underway to integrate MARL agents with the simulation environment.

Focus areas include:

- Interface between environment and agents
- Decision loop integration
- Reward feedback mechanisms

4. Preliminary Observations

From the baseline system:

- Fixed-time control performs adequately under uniform traffic
- Performance degrades under dynamic or uneven traffic conditions
- Queue buildup occurs at high-density intersections

These observations reinforce the need for **adaptive, learning-based control mechanisms**.

5. Challenges Identified

- Designing an effective reward function for traffic optimization
- Ensuring stable learning in a multi-agent environment
- Managing trade-offs between local and global optimization
- Computational constraints for scaling simulations

6. Next Steps

Short-Term

- Implement initial MARL agent model
- Integrate agent decision-making into simulation loop
- Run preliminary training experiments

Mid-Term

- Optimize learning performance
- Conduct comparative experiments with baseline
- Validate improvements across traffic scenarios

Long-Term

- Evaluate system under increased network complexity
- Introduce realistic traffic patterns
- Explore integration with real-world datasets

7. Summary

The project has successfully established a **robust experimental foundation**, including simulation, baseline system, and evaluation framework.

The next phase focuses on developing and validating a **Multi-Agent Reinforcement Learning-based adaptive system**, which represents the core contribution of this research.

Progress so far demonstrates **consistent advancement toward a scalable and intelligent traffic optimization framework**.